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Test Plan Documentation

Interactive House

Revision History

Date	Version	Description	Author
08/05/2026	1.0	Merged and updated test plan based on implemented system	Katerina Arvay, Dario Ostojic, Andre Sandblom

Test Plan

Test ID	Test Name	Priority
T1	User registration test	Essential
T2	User authentication test	Essential
T3	Device control test	Essential
T4	Device state synchronization test	Essential
T5	Unauthorized access test	Essential
T6	Database persistence test	Essential
T7	Concurrent request handling test	Desirable
T8	Gateway communication test	Essential
T9	Password management test	Essential

Test Cases

T1 – User registration test

Related requirement: R1 – Register user

Verifies that a new user can successfully register through the backend API. The system shall store user information persistently, hash the password for security, set up default device fields, and reject duplicate usernames.

Steps:

- Send a registration request with a valid username and password.
- Confirm the system responds with a success message.
- Check the database and verify the user was created with a hashed (not plain text) password.
- Verify that all 7 default device fields were created: white light, orange light, door, window, fan A, fan B, and buzzer.
- Try registering the same username again - the system should reject it.
- Try registering without the required API key - the system should reject it.

Expected result: New users are created successfully with secure password storage and all default devices. Duplicate usernames and missing API keys are rejected.

T2 – User authentication test

Related requirement: R2 - Authenticate user, R3 - Terminate session, R11 - Block unauthorized access

Verifies that users can log in with valid credentials and receive a session token. Invalid credentials, expired tokens, and missing tokens shall be rejected.

Steps:

- Log in with a valid username and password.
- Confirm a session token is returned.
- Try logging in with a wrong password - the system should reject it.
- Log out and confirm the token is no longer accepted.
- Try accessing a protected page without a token - the system should return an error.
- Use an expired token - the system should reject it.

Expected result: Login works with correct credentials. Wrong passwords, missing tokens, and expired tokens are all rejected appropriately.

T3 – Device control test

Related requirement: R4–R8, R15 - Device control and LCD display

Verifies that authenticated users can send commands to smart house devices through the backend API. The system shall update the device state and forward the command to the physical hardware via the gateway.

Steps:

- Log in and obtain a session token.
- Send a command to open the door - verify the state is saved and the door responds physically.
- Turn the white light on and off - verify both transitions are saved correctly.
- Set the orange light to 50% brightness - verify the value is stored correctly (range 0–255).
- Open and close the window - verify the gateway receives the correct command.
- Turn fan A and fan B on independently - verify each can be controlled on its own.
- Turn the buzzer on and off - verify it responds correctly.
- Send a short text message to the LCD display - verify it appears on screen (maximum 16 characters).

Expected result: All device commands are accepted, saved in the database, and forwarded to the physical hardware. The LCD display shows the correct text.

T4 – Device state synchronization test

Related requirement: R9 – Retrieve device status

Verifies that device states stored in the backend match what is shown to the user and reported by the gateway.

Steps:

- Set a device to a known state (for example, door open).
- Retrieve the current device status from the API.
- Confirm the returned status matches what was set.
- Confirm the password field is not included in the response.
- Check that the SG2 mobile app reflects the updated state in real time.

Expected result: The device status returned by the API always reflects the actual state. The password is never exposed in status responses. The mobile app shows the correct state.

T5 – Unauthorized access test

Related requirement: R11 - Block unauthorized access, S4 - Security

Verifies that requests without valid authentication are rejected and do not modify any system state.

Steps:

- Try sending a device command without being logged in - the system should reject it.
- Try using an invalid or expired token - the system should reject it.
- Confirm that no device state was changed after a rejected request.
- Check directly in the database that passwords are stored as hashed values, not plain text.

Expected result: All unauthenticated and unauthorized requests are rejected. No system state is modified. Passwords are always stored securely.

T6 – Database persistence test

Related requirement: R12 - Data persistence (partially implemented)

Verifies that critical system data survives a server restart or temporary disruption. Note: the secondary backup database (Supabase) is not yet implemented - this test covers the primary database only.

Steps:

- Set several devices to known states.
- Restart or simulate a disruption of the server.
- Reconnect and retrieve the device states.
- Confirm the last known states are preserved.
- Note: full backup and recovery testing is planned for a future release when the Supabase integration is complete.

Expected result: Device states and user data persist through server restarts. Full backup and recovery testing is planned for a future release.

T7 – Concurrent request handling test

Related requirement: S3 - Scalability (desirable)

Verifies that the backend can handle multiple simultaneous requests without causing inconsistent device states or data corruption.

Steps:

- Send multiple device commands from different sessions at the same time.
- Confirm that no requests are lost or cause errors.
- Verify that all device states are consistent after the test.

Expected result: The system handles concurrent requests without errors or inconsistency. This is a desirable test — not required for basic functionality.

T8 – Gateway communication test

Related requirement: R4–R8 - Device control, S8 - Deployment

Verifies that the backend correctly communicates with the SG3 gateway and that device commands are forwarded and acknowledged by the physical hardware.

Steps:

- Ensure the SG3 gateway is running and reachable via Cloudflare Tunnel.
- Send a device command through the API.
- Confirm the gateway receives and forwards the command to the Arduino.
- Verify the gateway health endpoint responds with OK.
- Run a full end-to-end test: register a user, log in, send a command, and verify the physical device responds.

Expected result: Commands pass correctly from the API through the gateway to the hardware. The gateway is reachable externally and passes its health check.

T9 – Password management test

Related requirement: R13 - Update password, R14 - Delete device field

Verifies that users can change their password and that custom device fields can be added and removed.

Steps:

- Log in with the current password.
- Submit a password change with the current and new password.
- Log in again with the new password - it should succeed.
- Try logging in with the old password - it should fail.
- Add a custom device field and then delete it - verify it is removed from the database.
- Try to delete the password field - the system should prevent this.

Expected result: Password changes take effect immediately. Old passwords stop working. Custom device fields can be deleted freely, but protected fields like the password cannot be removed.